MIL-S-20557
4 January 1952

MILITARY SPECIFICATION

SHELL, STEEL, FOR SMOOTH-BORE MORTARS,
MANUFACTURED FROM FORGINGS

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-37-1B
22 May 1945

This specification consists of this cover sheet, Specification 50-37-1B, dated 22 May 1945, and Amendment 2, dated 7 February 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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U. S. ARMY
SPECIFICATION

No. 50-37-1B
22 May 1945
Superseding
No. 50-37-1A
9 December 1943

SHELL, STEEL, FOR SMOOTH-BORE MORTARS, MANUFACTURED FROM FORGINGS

A. APPLICABLE SPECIFICATIONS

A-1. The Ordnance Department drawings pertaining to the shell under contract and all drawings and specifications listed or referred to thereon shall form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPE

B-1. This specification covers high explosive, chemical, and practice shell for smooth-bore mortars, manufactured from forgings.

C. MATERIAL AND WORKMANSHIP

C-1. Material and parts shall be in accordance with applicable specifications and drawings.

C-2. The shell shall be machined from forgings manufactured in accordance with applicable provisions of this specification.

C-3. The forgings shall be made from material manufactured in accordance with the steel specification prescribed on the forging drawing, or shell drawing, or in the contract or order.

C-4. Bars and billets shall be divided into forging blanks by any method desired by the contractor and which is satisfactory to the inspector. The contractor's process for dividing blanks must not develop injuries in a blank which may in turn result in a defective shell.

C-5. The machining and treatment of the shell shall be uniform so that test samples will be truly representative of the group from which they have been selected.

D. GENERAL REQUIREMENTS

D-1. All forgings, for which no heat treatment is required to meet the physical properties, shall be subjected to a uniform slow cooling process from the forging temperature down to not over 400°F. Forgings may be cooled at an accelerated rate from 400°F. to room temperature, provided sufficient checks are made with a contact pyrometer or by other suitable means to demonstrate to the satisfaction of the inspector that forgings have reached 400°F. or a lower temperature before being subjected to accelerated cooling. Forging manufacturers who are also machiners of finished shell may cool the forgings that they machine themselves at any rate which will produce the machinability they desire. Forgers who are teamed with machiners of finished shell may, upon agreement between themselves and such machiners, also cool their forgings as prescribed above. The cooling rate shall in no case result in defects such as cracks, ovalities of cavity or warping, and this fact must be proved to the satisfaction of the inspector.

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D-2. Shell requiring a nosing-in operation may be nosed-in either hot or cold. For shell requiring heat treatment to meet the specified physical properties, the nosing-in operation shall be made before the final heat treatment. No cold working involving plastic flow of metal is authorized after final heat treatment. Shell that require no heat treatment to meet the specified physical properties but which are cold nosed shall be stress-relieved after the cold-nosing operation by subjecting the shell to a minimum temperature of 700°F. for a minimum time of one-half hour.

D-3. All shell which require heat treatment to meet the specified physical properties shall, after the nosing-in operation, if such operation is required but before finish machining, be given a heat treatment wherein the shell is heated above the critical temperature range but not to exceed 1600°F. and quenched in oil and drawn without delay at a temperature suitable to produce required physical properties but not less than 750°F. After removal from the draw furnace shell shall be slowly cooled to a temperature of not more than 700°F., after which cooling may be accelerated by a water spray or similar means. The heat treatment shall be umiform throughout for each lot and shall be so conducted as to produce in any shell subsequently selected by the inspector the physical properties specified on the shell drawing.

E. DETAIL REQUIREMENTS

E-1. The shell shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Chemical composition. The chemical composition of the steel shall be that specified on the shell drawing or, if not specified on the shell drawing, it shall be that specified in the applicable steel specification, or as prescribed in the contract or order.

E-3. Physical properties. Unless otherwise specified on the shell drawing, the minimum physical properties determined from test coupons (see Par F-2) shall meet the following requirements.

Yield Point	35,000 pounds per square inch
Elongation in 2 inches	15 percent
Reduction of area	30 percent

E-4. The interior of all shell shall be smooth and free from burrs, scale, fins, deep scars, pits, and sharp edges.

E-5. All shell shall be sound and free from cracks and porosity.

E-6. Cleaning prior to painting. Upon completion of the machining operations the shell shall be cleaned to free them of all grease, dirt, chips, and other foreign material, preparatory to painting. The shell surfaces to be painted shall be cleaned by one of the methods covered by Specification AXS-1245. These cleaned surfaces shall not be recontaminated prior to painting. At the time of painting these surfaces shall meet the various performance requirements of Specification AXS-1245. The interior cavity of each shell shall be cleaned by the shot or sand blast method, or by any other method approved by the Chief of Ordnance.

(50-37-1B)

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E-7. Coating. The exterior coating and the cavity coating specified on the drawing for the shell shall be applied in such a manner as to produce a tightly adherent coating of uniform thickness over the entire specified areas. The shell shall be stencilled or marked as prescribed on the drawing. All paint and marking shall be dry to the touch before shell are packed.

F. METHOD OF SAMPLING, INSPECTION AND TESTS

F-1. Inspection. Inspection shall be as prescribed in U. S. Army Specification No. 50-0-1, "General Specifications for Ammunition Except Small Arms Ammunition", and Federal Specification No. QQ-M-151, "Metals, General Specification for Inspection of".

F-2. Forgings.

F-2a. A lot of forgings shall consist of all the forgings made from one heat or melt of steel and treated by the same method.

F-2b. A smaller quantity may constitute a lot at the discretion of the inspector.

F-2c. The first lot manufactured on any contract shall not exceed 5,000 shell, except when a new contract immediately follows the manufacture of satisfactory material of the same character.

F-2d. Tension test, selection of coupons. For lots of 2,000 forgings or less, two coupons shall be taken from at least one finished forging of each 1,000 or fraction thereof. For lots of more than 2,000 forgings, coupons shall be taken from two finished forgings of each 10,000 or fraction thereof. Two test coupons equal in width to three times the wall thickness of the forging shall be cut from each forging selected for test. The test coupons shall be taken longitudinally from the diametrically opposite sides of the wall of each of the sample forgings. The end of the coupons shall be located approximately in line with the bottom of the cavity.

F-2e. Tension test, test specimens. One tension test specimen shall be machined from the center of each test coupon. Tension test coupons shall conform to type 1, Federal Specification QQ-M-151, or (if the wall thickness of the forging is not sufficient for type 1) to the largest size of type 4 specimen obtainable.

F-2f. Retests. If any of the test specimens representing a lot of forgings fails to conform to the physical requirements, two additional test specimens from each forging that failed shall be selected by the inspector to represent the lot, and each shall be tested and shall show the required physical properties. In case of failure on retest, two replacing forgings may be selected by the inspector to represent each forging that failed in retest and tested in the same manner as the original test. All replacing test specimens shall meet the specified requirements to pass the lot, Any lot of forgings that fails to meet the physical requirements may be heat-treated or reheat-treated and resubmitted for test in the same manner as an original lot.

F-2g. A report submitted by the steel manufacturer, together with the forger's report of tension tests and details of heat treatment employed for each heat, shall be furnished the consignee by the manufacturer with invoice. These reports shall embody and results obtained by the steel maker and the forger for the use of the machiner.

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F-2h. Reports on standard forms prescribed by the Ordnance Department covering chemical composition, tension tests, and details of heat-treatment employed for each heat, shall be forwarded to the Ordnance Department by the Inspector.

F-3. Shell.

F-3a. A lot of shell shall consist of all shell made from one lot of forgings and treated by the same method. Where normal production lots of forgings do not exceed 5000 forgings, shell machined from such lots may be recoded or regrouped into larger lots not exceeding 20,000 shell provided that such lots shall represent forgings produced by one process and treated by the same method by one manufacturer at one plant. In order to avoid many small lots, shell left over from their original lots may be grouped together into miscellaneous lots, the sizes of each of which shall not exceed 5,000 shell.

F-3b. A smaller quantity may constitute a lot at the discretion of the inspector.

F-3c. The first lot manufactured on any contract shall not exceed 5,000 shell, except when a new contract immediately follows the manufacture of satisfactory material of the same character.

F-3d. Each shell shall be inspected for smoothness of cavidy. All shell showing burrs, scale, fins, deep scars, pits or sharp edges shall be rejected.

F-3e. Each shell shall be inspected for cracks and porosity. All shell which show cracks or porosity shall be rejected. Doubtful shell shall be referred to the inspector for disposition. (See paragraph F-4).

F-3f. The inspector shall assure himself that the operations of cleaning the cavity and exterior of the shell are performed in such a manner as to remove all foreign material and to insure that the surfaces meet the various performance requirements of Specification AXS-1245.

F-3g. Each shell shall be inspected to determine that the interior and exterior coatings have been evenly and thoroughly applied. Also, that the stencilling and marking is as prescribed on the drawing.

F-3h. Repairs and retests. Shell showing imperfections may be repaired by arc welding (shielded arc) or gas welding provided the requirements listed below are met. No single imperfection exceeding 1/2-inch in length shall be repaired and no more than two such imperfections shall be repaired on a shell. The imperfection shall be completely removed before welding and a weld of 100% penetration shall be required. However, if the imperfection is less than 1/3 the thickness of the shell, a weld of 100% penetration will not be necessary. The repaired shell shall then be tested at 4000 pounds per square inch hydraulic pressure. Any shell failing on this retest shall be finally rejected.

F-4. There shall be installed in each plant engaged in machining shell one set of hydraulic equipment capable of testing shell by an internal hydrostatic pressure to destruction. The equipment shall be used by the inspector for testing doubtful shell, such as those which show evidence of cracks or porosity.

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F-5. In determining the hydrostatic pressure to be used for stressing the metal of the body up to the prescribed yield point of the material, the following formula shall be used.

$$P = \frac{2\,St}{D}$$

where

P = pressure in pounds per square inch.
S = yield point of metal
t = thickness of wall at weakest section.
D = outside diameter at weakest section.

F-6. If the contractor elects to scrap doubtful shell rather than submit them to hydrostatic pressure test, the hydraulic equipment may be omitted.

G. PACKAGING, PACKING AND MARKING

G-1. Packing.

G-1a. Shell shall be packed in commercial type containers so constructed as to insure acceptance by common or other carriers, for safe transportation at the lowest rate to the point of delivery.

G-1b. Data card. Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted shell in accordance with the form specified below. Two of these cards completely filled out for each lot shall accompany each shipping ticket and six cards completely filled out shall be furnished the inspector. Lot number of contents shall be marked plainly on each container.

AMMUNITION DATA CARD

Kind_____ Ammunition Lot No._____
Quantity in lot_____
Drg. No._____ Rev._____ Spec. No._____ Rev._____
Manufactured by_____ Date_____
Contract No._____ Order No._____
Forging Manufacturer_____ Heat No._____
Packed_____ in a box
Remarks_____
Certified to by_____ Inspector
(Print here name of inspector, district, arsenal, or loading plant, and sub-district where applicable.)

G-2. Marking. The marking for shipment shall be in accordance with U. S. Army Standard Specifications for Marking Shipments by Contractors, 100-2.

H. NOTES

H-1. The shell covered by this specification are for use in Smooth-Bore Mortars.

H-2. The use of this specification whenever applicable, is mandatory on all procuring agencies of the Army.

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Note: Copies of this specification may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington, D. C.

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U. S. ARMY
SPECIFICATION

No. 50-37-1B
Amendment 2
7 February 1949
Superseding
Amendment 1

SHELL, STEEL, FOR SMOOTH-BORE MORTARS, MANUFACTURED FROM FORGINGS

The following changes shall be made in Specification No. 50-37-1B, dated 22 May 1945:

Page 2, paragraph D-2. Delete the last sentence and substitute the following:

"All shell not requiring an adapter and not requiring heat treatment to meet the specified physical properties but which are cold nosed shall be stress-relieved after the cold nosing operation by subjecting the shell to a minimum temperature of 700°F. for a minimum time of one-half hour."

* Page 2, paragraph E-4, add the following sentence at eht end of the paragraph:

"The interior cavity of each shell shall be cleaned by the shot or sand blast method, or by any other method approved by the Chief of Ordnance."

* Page 2, paragraph E-6, delete the paragraph and substitute the following:

"E-6. Cleaning Priot To Painting.--The shell surfaces to be painted shall be cleaned and prepared in accordance with Grade I, Specification JAN-C-490. These cleaned surfaces shall not be recontaminated prior to painting. After painting, these surfaces shall meet the various performance requirements of Specification JAN-C-490."

* Page 4, paragraph F-3f, delete "Specification AXS-1245" and substitute "Specification JAN-C-490".

* Asterisks indicate changes in addition to or differing from pervious amendments.

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